

KNOWLEDGE GRAPH ASSIGNMENTS

ASSIGNMENT 1 – KNOWLEDGE GRAPH CREATION AND QUERYING USING CYPHER QUERIES

TASKS

1) Take a hospital/medical setup containing doctors, nurses, patients, diseases, treatments and similar entities. Create nodes, edges/relationships by using synthetic data using Python and neo4j driver.

Take help of any AI Chatbot to generate the cypher queries to create them

2) Verify the graph is created successfully in the neo4j dashboard > Explorer

3) Write at least three cypher queries to demonstrate the usage. For example:

- 1) Which doctors can treat diabetes?
- 2) Which medications are commonly prescribed for hypertension? Which one is most popular?
- 3) How total patients do we have?

Submit: colab notebook and screenshot of neo4j graph from the explorer (you can print the graph in colab itself or submit separately)

Put the links to above in a document inside “knowledgegraph” folder in ur top level submissions folder.

ASSIGNMENT 2 – LLM FOR CYPHER QUERY AND RESPONSE GENERATION

Extend the code from assignment 1 to do the following:

Simulate a retrieval process where a prompt is the input and use LLM to

- a) Generate the cypher query from the prompt
- b) generate the final response to the user question (prompt)

Submission

- 1) Working colab notebook (share link) or .py
- 2) Screenshots of prompt and response

Submit this by putting a word document with links inside the “knowledgegraph” folder inside your shared submissions folder